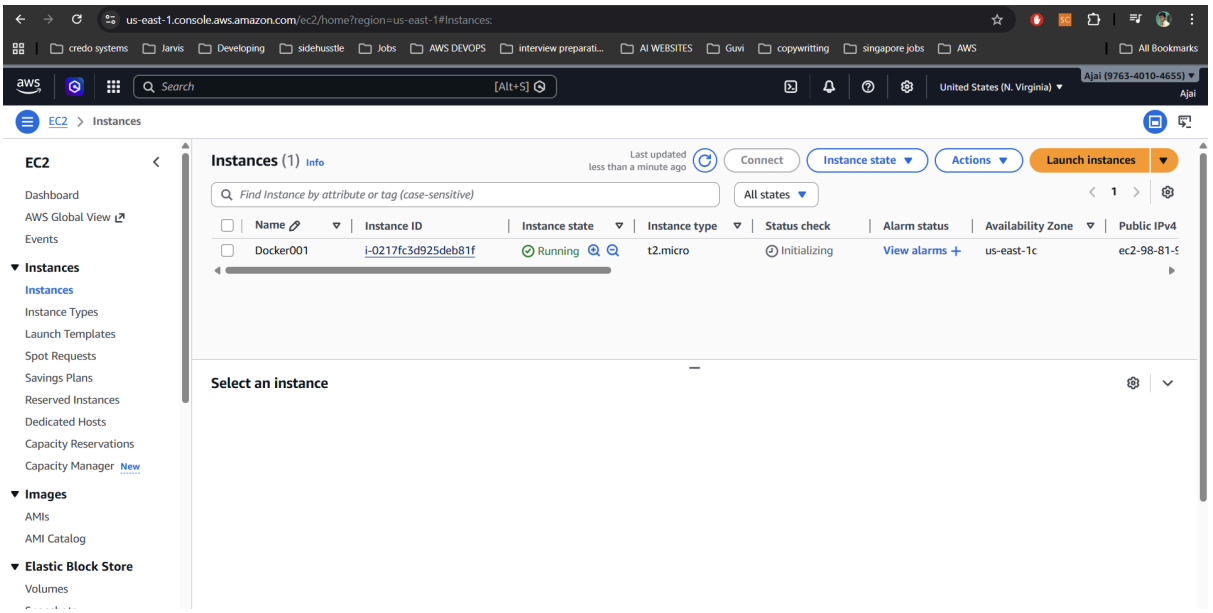
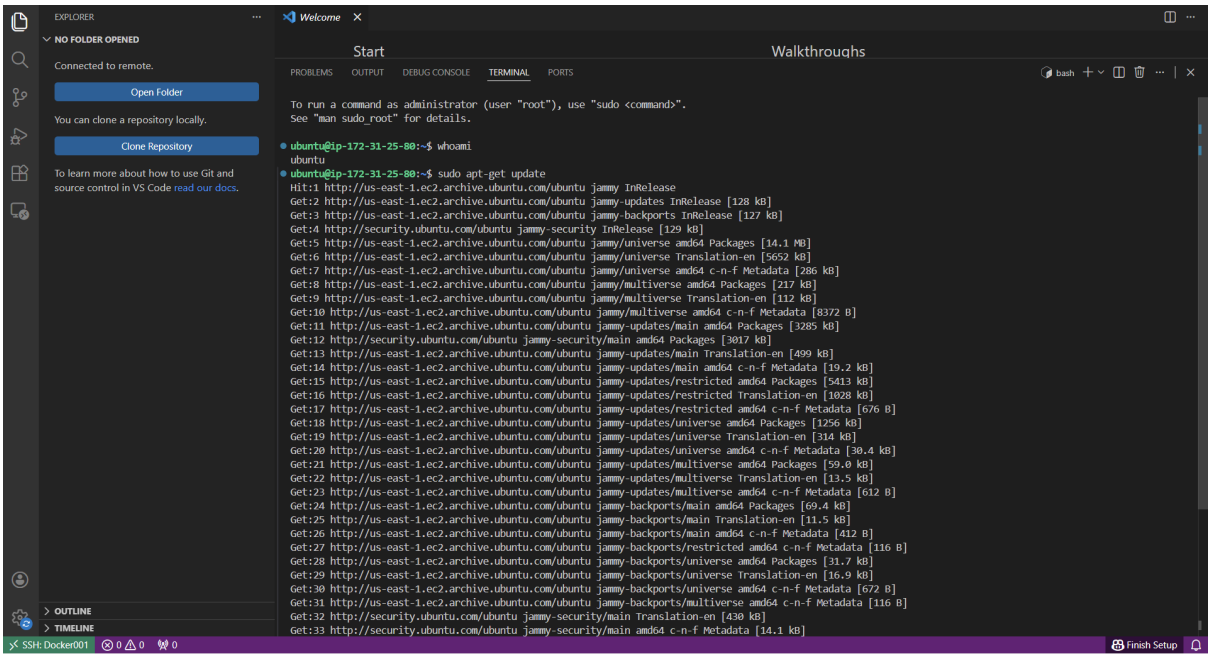
Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network)



I will open Visual Studio Code and configure the necessary tools to run Docker commands. After establishing an SSH connection to the remote server, I will proceed to update the server using the appropriate command.

sudo apt-get update



And next i will install docker using this command: **sudo apt-get install -y** docker.io
To check the version **docker --version**

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                                bash

ubuntu@ip-172-31-25-80:~$ sudo apt-get install -y docker.io
Unpacking dnsmasq-base (2.90-0ubuntu0.22.04.1) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../6-docker.io_28.2.2-0ubuntu1~22.04.1_amd64.deb ...
Unpacking docker.io (28.2.2-0ubuntu1~22.04.1) ...
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../7-ubuntu-fan_0.12.16_all.deb ...
Unpacking ubuntu-fan (0.12.16) ...
Setting up dnsmasq-base (2.90-0ubuntu0.22.04.1) ...
Setting up runc (1.3.3-0ubuntu1~22.04.3) ...
Setting up dns-root-data (2024071801~ubuntu0.22.04.1) ...
Setting up bridge-utils (1.7-1ubuntu3) ...
Setting up pigz (2.6-1) ...
Setting up containerd (1.7.28-0ubuntu1~22.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /lib/systemd/system/containerd.service.
Setting up ubuntu-fan (0.12.16) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /lib/systemd/system/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1~22.04.1) ...
Adding group `docker' (GID 122) ...
Done.
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /lib/systemd/system/docker.socket.
Processing triggers for dbus (1.12.20-2ubuntu4.1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-25-80:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1
ubuntu@ip-172-31-25-80:~$
```

To check the status, we can either use a two commands

sudo service status docker(or)
sudo systemctl status docker

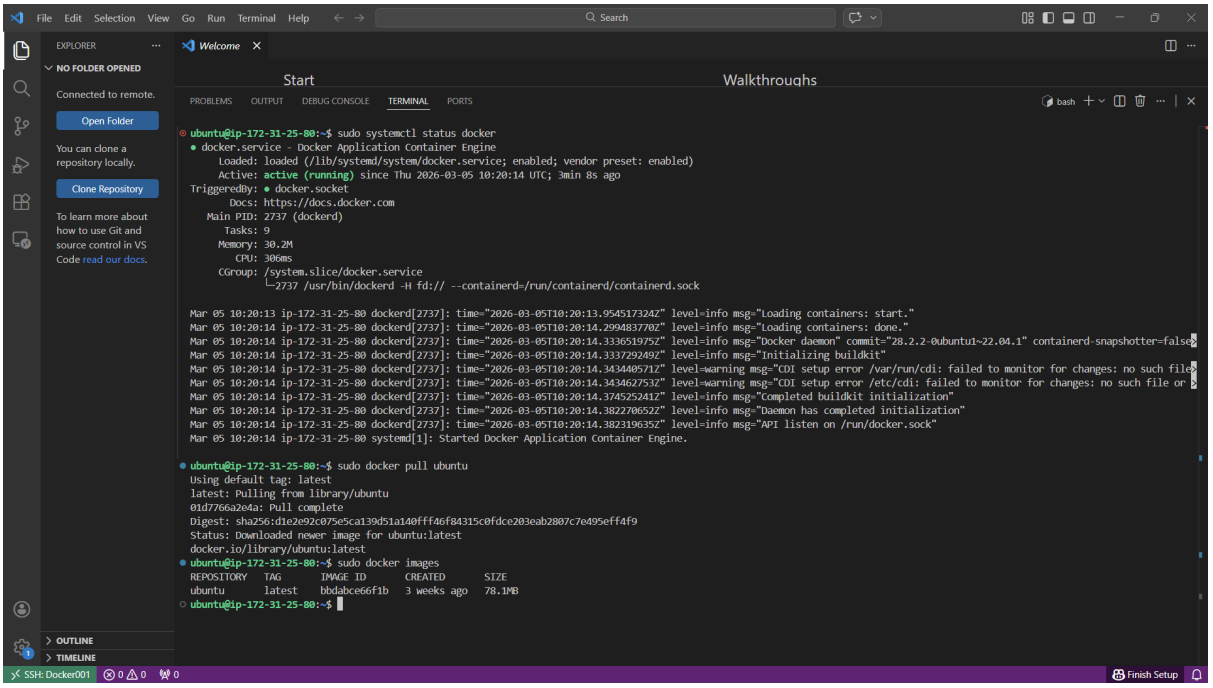
```
File  Edit  Selection  View  Go  Run  Terminal  Help                    Search
EXPLORER                                  Welcome
NO FOLDER OPENED
Connected to remote.                      Start                          Walkthroughs
Open Folder
You can clone a repository locally.
Clone Repository
To learn more about how to use Git and source control in VS Code read our docs.

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                                sudo

ubuntu@ip-172-31-25-80:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/lib/systemd/system/docker.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-03-05 10:20:14 UTC; 3min 8s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2737 (dockerd)
      Tasks: 9
     Memory: 30.2M
        CPU: 306ms
     CGroup: /system.slice/docker.service
             └─2737 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Mar 05 10:20:13 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:13.954517324Z" level=info msg="Loading containers: start."
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.299483770Z" level=info msg="Loading containers: done."
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.333651975Z" level=info msg="Docker daemon" commit="28.2.2-0ubuntu1~22.04.1" containerd-snapshotter=false
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.333729249Z" level=info msg="Initializing buildkit"
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.343440571Z" level=warning msg="CDI setup error /var/run/cdi: failed to monitor for changes: no such file
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.343462753Z" level=warning msg="CDI setup error /etc/cdi: failed to monitor for changes: no such file or
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.374525241Z" level=info msg="Completed buildkit initialization"
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.382270652Z" level=info msg="Daemon has completed initialization"
Mar 05 10:20:14 ip-172-31-25-80 dockerd[2737]: time="2026-03-05T10:20:14.382319635Z" level=info msg="API listen on /run/docker.sock"
Mar 05 10:20:14 ip-172-31-25-80 systemd[1]: Started Docker Application Container Engine.
lines 1-22/22 (END)
```

Docker- Images

I will now execute operations using Docker commands, specifically by pulling images from Docker Hub.

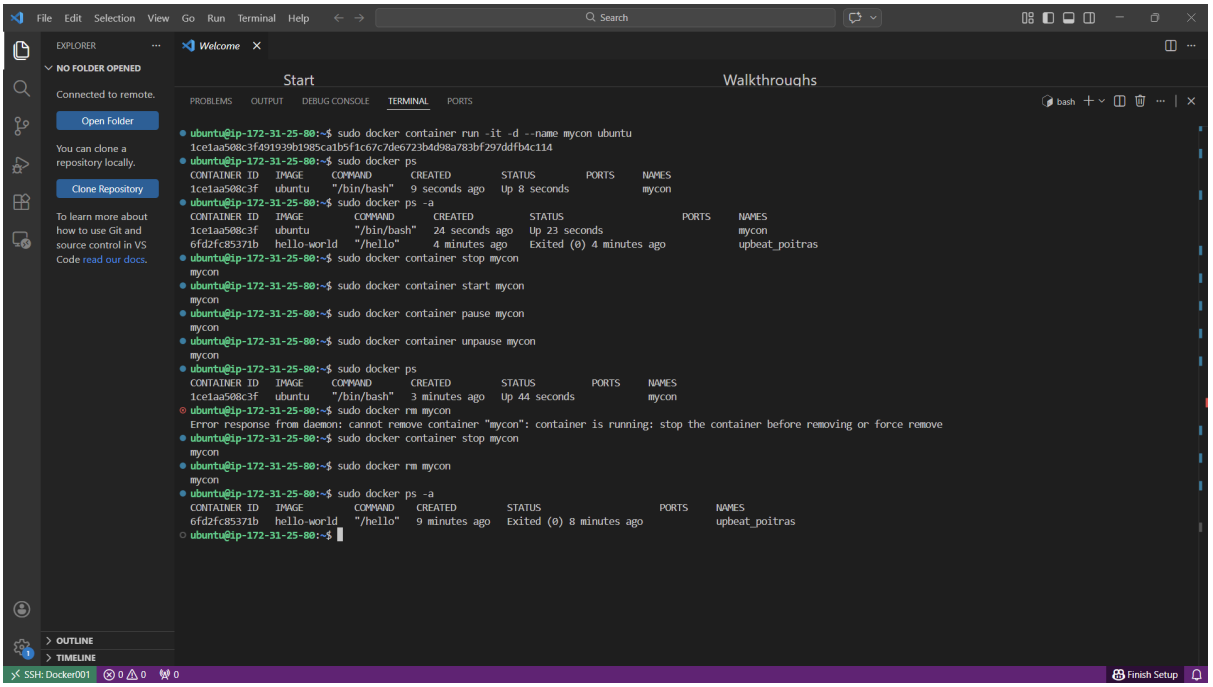
Sudo docker pull ubuntu

Sudo docker images



The command **`sudo docker container run -it -d –name mycon ubuntu`** is used to create and start a new Docker container from the Ubuntu image. The options `-it` allow interactive terminal access, `-d` runs the container in the background (detached mode), and `--name mycon` assigns the container a custom name mycon. The `sudo` gives administrator permission to execute the Docker command.

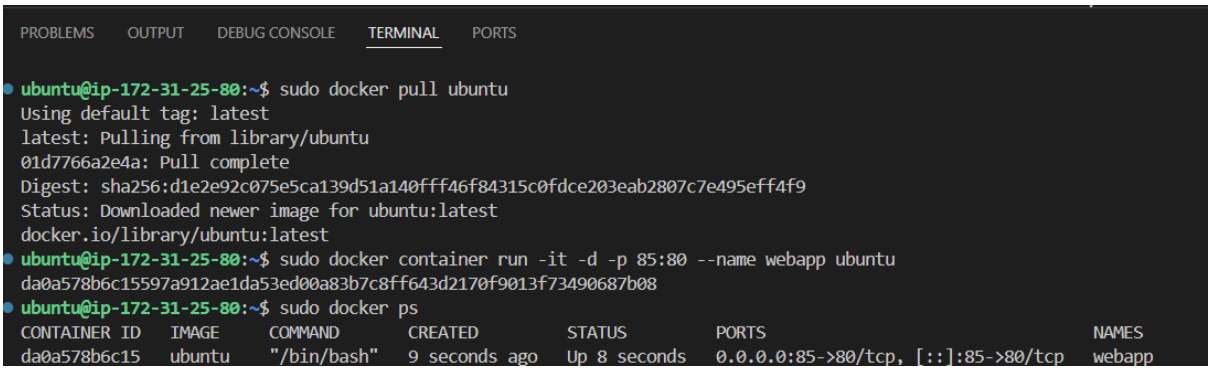
sudo docker ps
sudo docker ps -a
sudo docker container stop mycon
sudo docker container start mycon
sudo docker container pause mycon
sudo docker container unpause mycon
sudo docker ps
sudo docker rm mycon
sudo docker container stop mycon
sudo docker rm mycon
sudo docker ps -a



Docker - containers

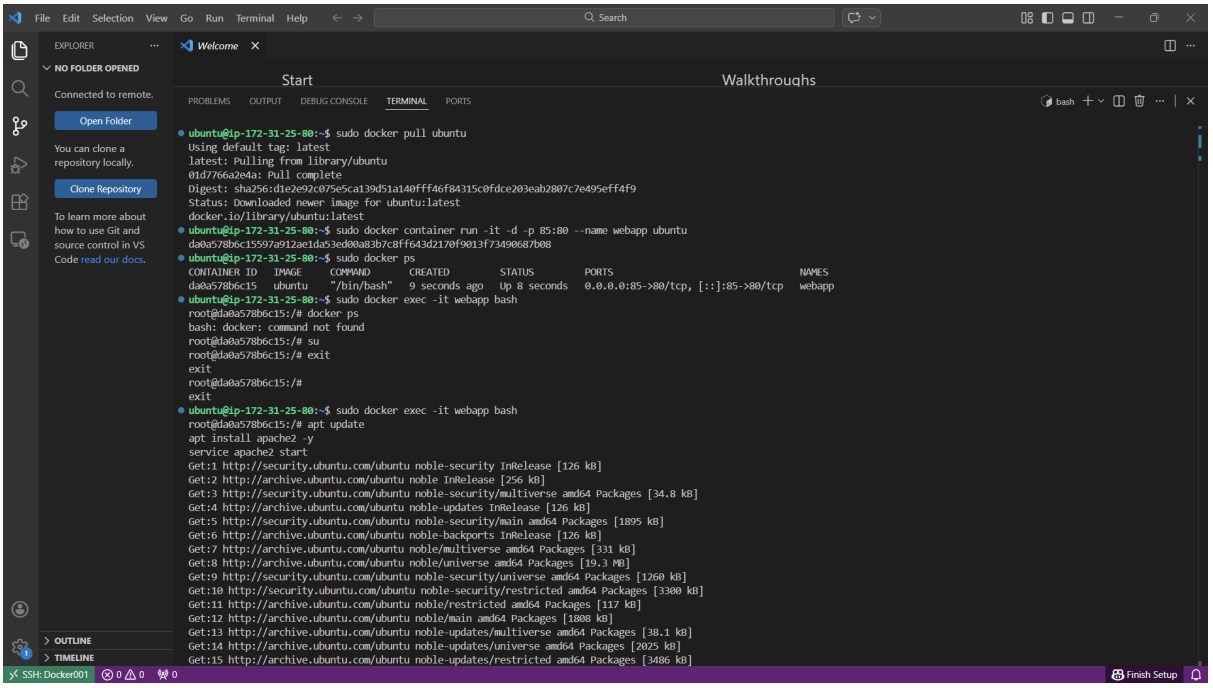
The command **`sudo docker container run -it -d -p 85:80 --name webapp`** `ubuntu` is used to create and start a Docker container named `webapp` from the Ubuntu image. The options `-it` allow interactive terminal support, `-d` runs the container in the background, and `-p 85:80` maps port 85 of the host machine to port 80 of the container, allowing external access to services running inside the container.

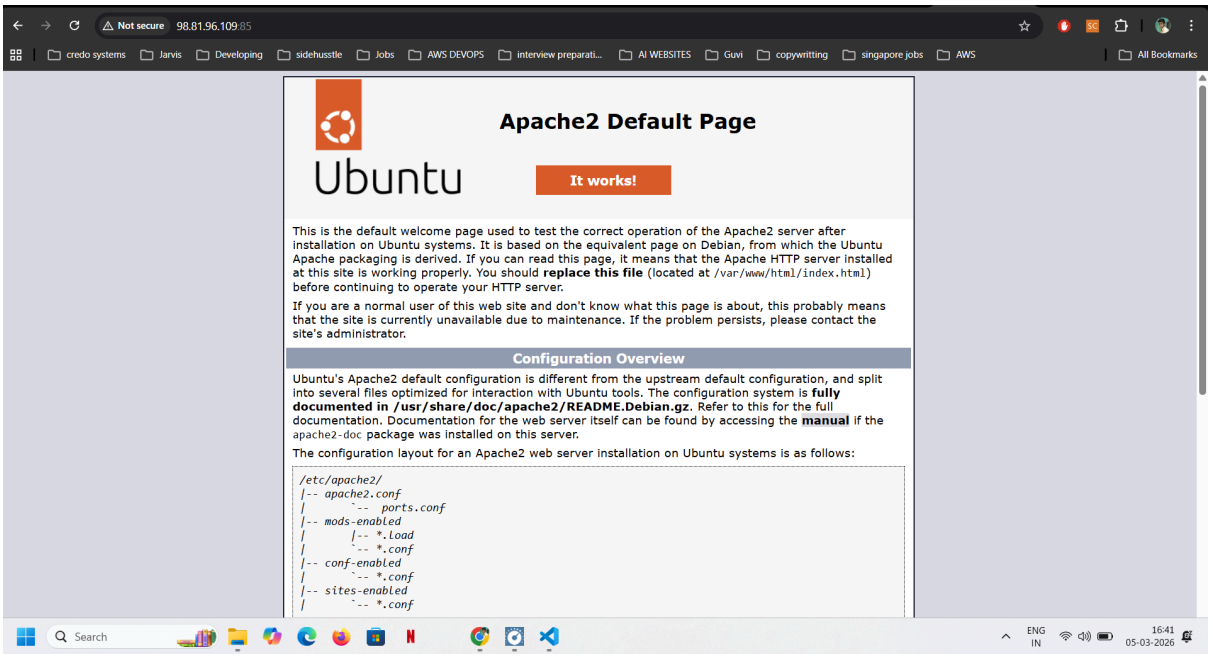
The command `sudo docker exec -it webapp bash` is used to enter the running container named `webapp` and open a bash shell inside it, allowing you to interact with the container's environment and execute commands directly within it.



Since the **Ubuntu image does not contain any web server** by default. So i will install a apache2 server in a docker container now i am going to start a apache2 using this command

Sudo exec -it webapp service apache2 start





Docker volume and network

```
ubuntu@ip-172-31-25-80:~$ docker volume create myvolume
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/volumes/create": dial unix /var/run/docker.sock: connect: permission denied
ubuntu@ip-172-31-25-80:~$ sudo docker volume create myvolume
myvolume
ubuntu@ip-172-31-25-80:~$ sudo docker volume ls
DRIVER    VOLUME NAME
local     myvolume
ubuntu@ip-172-31-25-80:~$ sudo docker volume inspect myvolume
[
    {
        "CreatedAt": "2026-03-05T11:15:33Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/myvolume/_data",
        "Name": "myvolume",
        "Options": null,
        "Scope": "local"
    }
]
ubuntu@ip-172-31-25-80:~$
```

Network

```
ubuntu@ip-172-31-25-80:~$ sudo docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
1f4688b0e9e0   bridge    bridge    local
279e926437cb   host      host      local
1947cf6f4ac2   none      null      local
ubuntu@ip-172-31-25-80:~$ sudo docker network create my network
docker: 'docker network create' requires 1 argument

Usage:  docker network create [OPTIONS] NETWORK

Run 'docker network create --help' for more information
ubuntu@ip-172-31-25-80:~$ sudo docker network create mynetwork
2d6de7bae63b7dea2073d956be3a70dc12cb8fd7e63ab04fdaa00955b483b7dc
ubuntu@ip-172-31-25-80:~$ sudo docker run -d --network=mynetwork nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
206356c42440: Pull complete
b47f187216b6: Pull complete
1ad233904a11: Pull complete
eedda9fd8786: Pull complete
35ff83c394d6: Pull complete
17d0911eaf62: Pull complete
df0b66c867e4: Pull complete
Digest: sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
Status: Downloaded newer image for nginx:latest
7c4a70023f224bff372d33816051a063cc59e16657c23cd0094ada76246df4a2
ubuntu@ip-172-31-25-80:~$ sudo docker network inspect mynetwork
[
    {
        "Name": "mynetwork",
        "Id": "2d6de7bae63b7dea2073d956be3a70dc12cb8fd7e63ab04fdaa00955b483b7dc",
        "Created": "2026-03-05T11:18:14.7427464662",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
```